Roll No................

End Semester Examination

Name of the Program: B.Tech (ECE) Semester: V

Name of the Course: Communication Systems II Course Code: TEC 502

Time: 3 Hours Maximum Marks: 100

Note:

(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question

(iii) Total marks in each main question are twenty

(iv) Each question carries 10 marks.

Q1	(20 marks)	**CO1**
(a)	State and prove the sampling theorem for lowpass signal.	
(b)	Explain Delta Modulation (DM) in detail with suitable diagram.	
(c)	(i) In a binary PCM system, the output signal to quantization noise ratio is to be held to a minimum of 40 dB. Determine the number of quantization levels, and find the corresponding output signal to quantization noise ratio. (ii) A signal $x_1(t)$ is bandlimited to 2 KHz while $x_2(t)$ is bandlimited to 3 KHz. Find the Nyquist rate for (a) $x_1(2t)$ (b) $x_2(t-3)$ (c) $x_1(t)+ x_2(t)$.	
Q2	(20 marks)	**CO 2**
(a)	Determine the power spectral density for the Unipolar NRZ format.	
(b)	Explain the Nyquist criterion for distortionless baseband binary data transmission.	
(c)	How does the frequency-domain shape of a raised-cosine filter look, and how does roll-off factor (α) influence it?	
Q3	(20 marks)	**CO2 & 3**
(a)	Give detail proof of Gram-schmidt orthogonalization procedure.	
(b)	Derive an expression for the probability of error (P_e) for the optimum filter in the simplest binary case.	
(c)	(i) Let $g(t)=e^{-\pi t^2}$, and h(t) is a filter matched to $g(t)$. If $g(t)$ is applied as input to h(t), Find the Fourier transform of the output. (ii) A communication channel of bandwidth 75 KHz is required to transmit binary data at a rate of 0.1 Mbps using raised cosine pulses. Determine the roll off factor α.	
Q4	(20 marks)	**CO4**
(a)	Derive an expression for probability of error of BPSK wave.	
(b)	Draw the block diagram of BFSK system and explain its working.	
(c)	Represent BASK signals in the signal space and find the Euclidean distance.	
Q5	(20 marks)	
(a)	Draw the block diagram of a digital communication system and explain the function of each block.	

(b)	A DMS X has four symbols x_1, x_2, x_3, and x_4 with $P(x_1)=1/2$, $P(x_2)=1/4$, and $P(x_3)=P(x_4)=1/8$. (i) Construct a shannon-Fano code for X, and calculate the efficiency of the code. (ii) Repeat for the Huffman code and compare the results.	**CO5 & CO6**
(c)	Prove that the upper bound on the value of entropy $H(X)$ of a source X is $\log_2 M$ where M is the size of the alphabet of X.	